
Assignment 2 - Linear Regression and Classification

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1 Abstract

This report presents an exploration of linear and logistic regression models applied to classification tasks. We went through the full experimental pipeline—from data acquisition and preprocessing to model implementation and evaluation—on two benchmark datasets: Breast Cancer and Wine Recognition. We found that the logistic and multiclass regression approaches achieved marginally better classification accuracy and robustness compared to linear regression, demonstrating their effectiveness for these tasks.

2 Introduction

In modern machine learning, selecting the appropriate model is crucial for building robust classifiers. This project investigates both regression and classification approaches by implementing linear regression, logistic regression, and multiclass classification from scratch. We applied these methods to two widely studied datasets: the Breast Cancer dataset, containing 569 samples with 30 continuous features, and the Wine Recognition dataset, comprising 178 samples with 13 continuous features. Our experimental pipeline included data acquisition, preprocessing, feature analysis, the development of multiple linear regression models and logistic regression models for binary and multiclass tasks. Ultimately, our results indicate that while the performance of linear regression is close, logistic regression yields more reliable probabilistic predictions, resulting in slightly improved classification outcomes.

3 Datasets

In our experiments, we use two benchmark datasets from the UCI Machine Learning Repository [1, 2]:

For the Breast Cancer Wisconsin (Diagnostic) dataset, there are 569 samples, each representing a fine needle aspirate of a breast mass. The dataset contains 30 continuous features describing various cell nucleus characteristics (radius, texture, perimeter, area, smoothness, etc.), and the target variable indicates whether the tumor is malignant (1) or benign (0). There are no missing values, and the data are well-studied for classification tasks.

For the Wine Recognition dataset, there are 178 samples with 13 features derived from chemical analyses of wines grown in the same region in Italy. The three wine cultivars are represented by labels 1, 2, and 3. We use one-hot encoding for these labels in our multiclass experiments. As with the breast cancer data, there are no missing values in the wine dataset as well.

After splitting each dataset into training, validation, and test sets (70%-15%-15%), we standardize the features (subtract mean, divide by standard deviation) using statistics from each set separately. This prevents data leakage and ensures that our validation and test results are not artificially inflated. For the breast cancer data, we encode the targets as 0 for benign and 1 for malignant. For the wine data, we convert the class labels into one-hot vectors.

To gain an initial sense of which features correlate most strongly with each target, we compute simple regression coefficients by multiplying the transpose of the (standardized) feature matrix by the target vector (or matrix) and dividing by the number of training samples. For the breast cancer dataset, this yields a single coefficient per feature, indicating how strongly it aligns with the malignant diagnosis. For the wine dataset, we obtain a coefficient matrix of size (*number of features* $\times$ 3), reflecting how each feature contributes to each of the three wine classes.

In the breast cancer dataset, features related to tumor size and shape irregularity (e.g., `perimeter3`, `concave.points3`, `radius3`) show high positive coefficients, suggesting a strong correlation with malignant diagnoses. On the other hand, features such as `fractional.dimension2` or `symmetry2` tend to have lower or

even negative coefficients, indicating a weaker or inverse correlation with malignancy. These observations align with medical literature [2] that emphasizes irregular shapes and larger sizes as key indicators of malignancy.

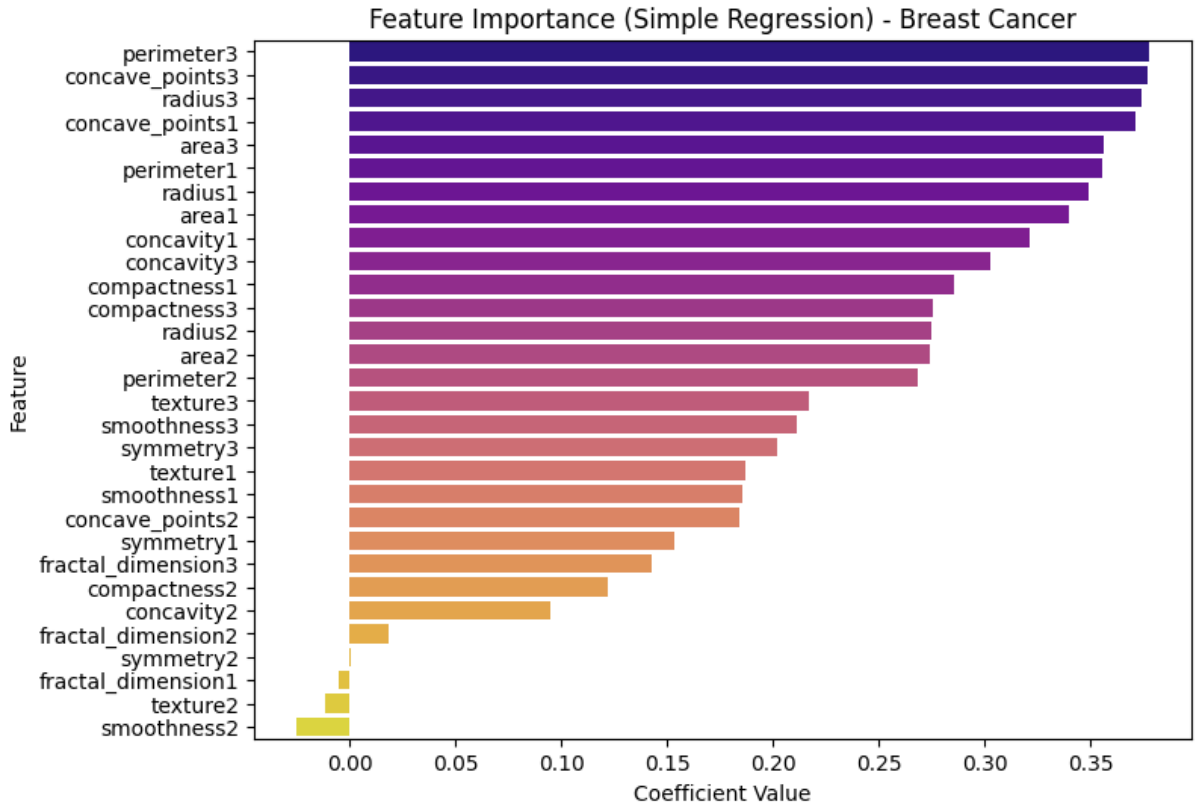
In the wine dataset, features like Alcohol, Color_intensity, and Malicacid often emerge as influential for distinguishing among the three cultivars. For instance, class_1 shows strong positive coefficients for Proline, Alcohol and Flavanoids, while class_3 is more influenced by Malicacid and Color_intensity. Prior studies also highlight phenolic compounds (Flavanoids, Nonflavanoid_phenols) and color-related measurements (Hue, Color_intensity) as important for differentiating wine types.

By examining these simple regression coefficients, we obtain a quick and intuitive ranking of features for both binary and multiclass tasks. These insights help guide more advanced modeling steps, such as logistic regression or regularized methods, where the interplay among features can be captured more effectively.

4 Results

Below are the results doe the breast cancer dataset:

1. Horizontal bar plots from the Simple Linear Regression in Task 1 on the datasets, with the coefficient weight as the x-axis and the feature names as the y-axis.



Figuur 1: Bar Plot for feature importance for breast cancer dataset

2. The difference between analytical and numerical gradients for logistic regression is 1×10^{-10} .
3. The convergence plot of the Cross Entropy loss function over iterations is as follows:

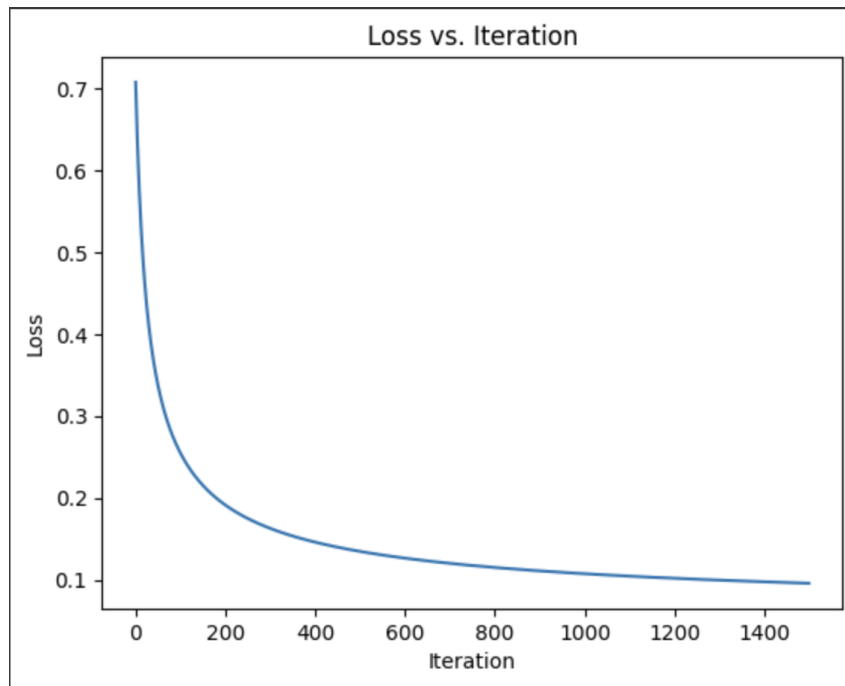
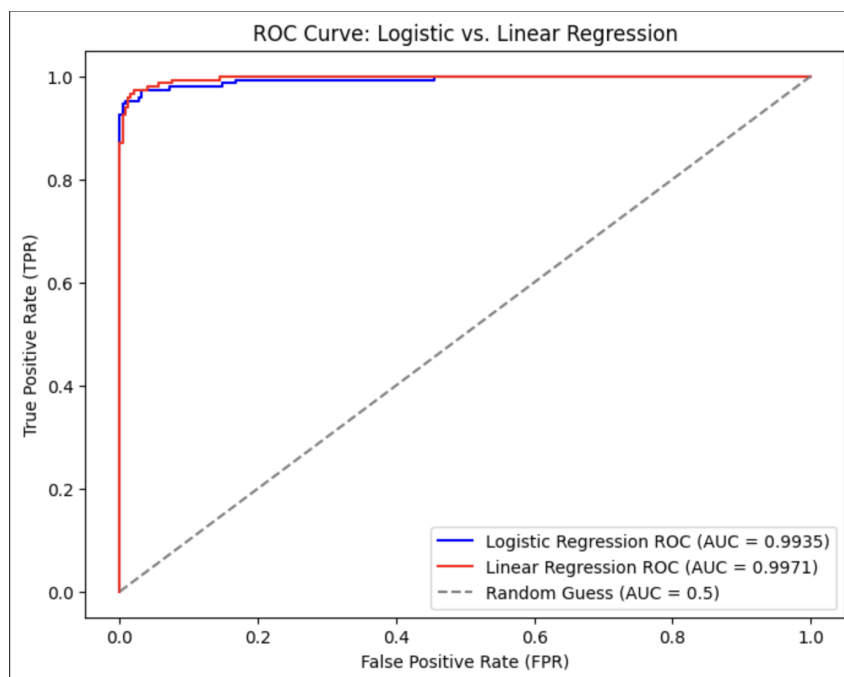


Figure 2: Convergence of the Cross Entropy loss

4. Comparison of ROC curves of logistic regression and multiple linear regression is as follows:



6. Bar plot of coefficients and features is as follows:

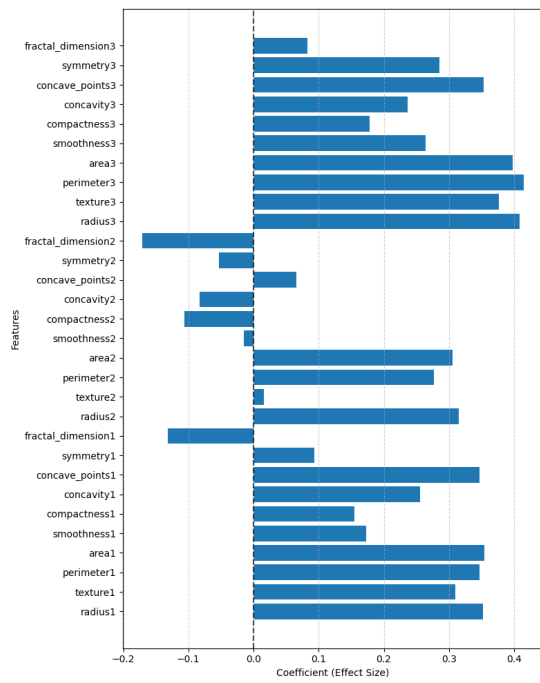


Figure 3: Bar plot of coefficients for logistic regression

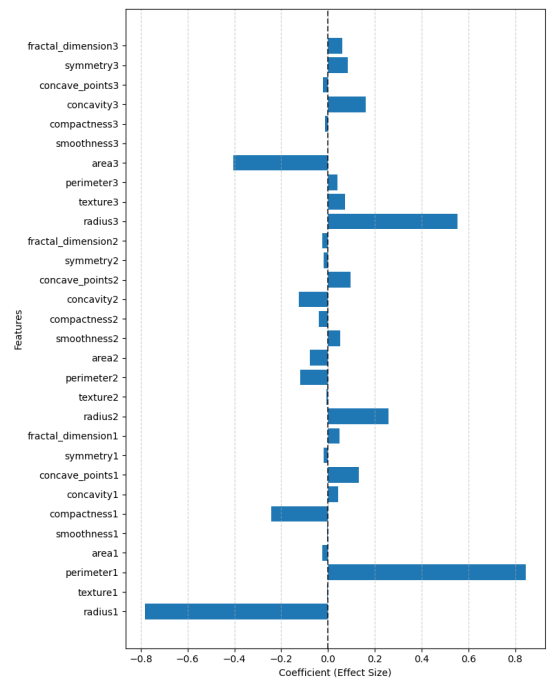


Figure 4: Bar plot of the coefficients for linear regression

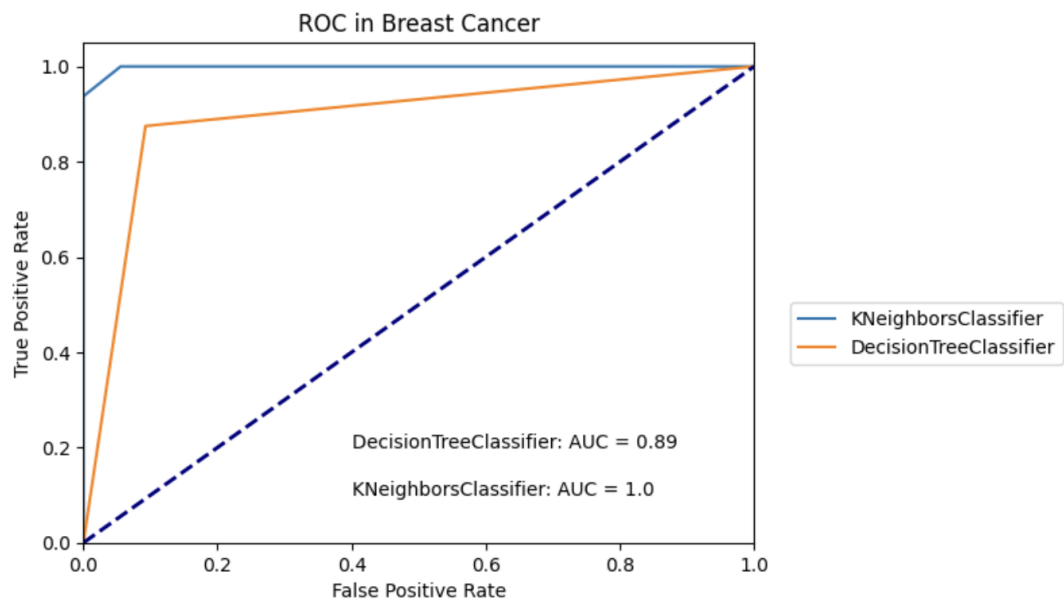


Figure 5: KNN and DT AUROC for Breast Cancer data

And below are the results for the multi-class wine dataset:

1. Horizontal bar plots from the Simple Linear Regression in Task 1 on the datasets, with the coefficient weight as the x-axis and the feature names as the y-axis.

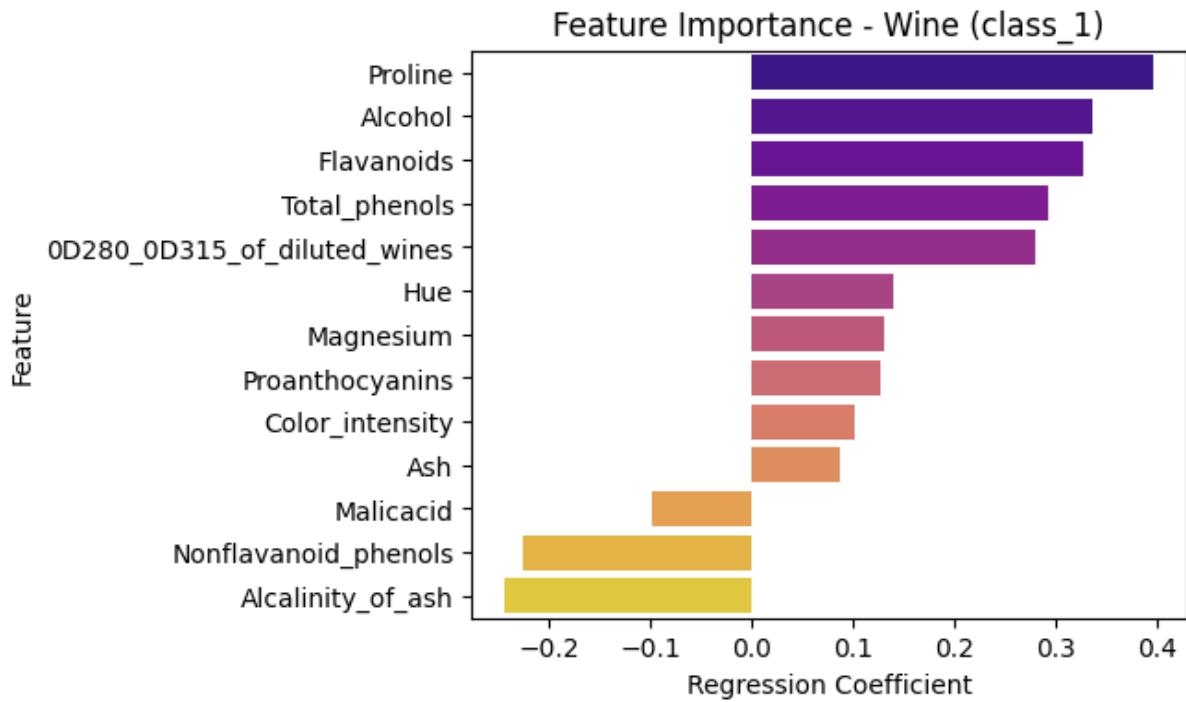


Figure 6: Bar Plot for feature importance for class 1 of wine dataset

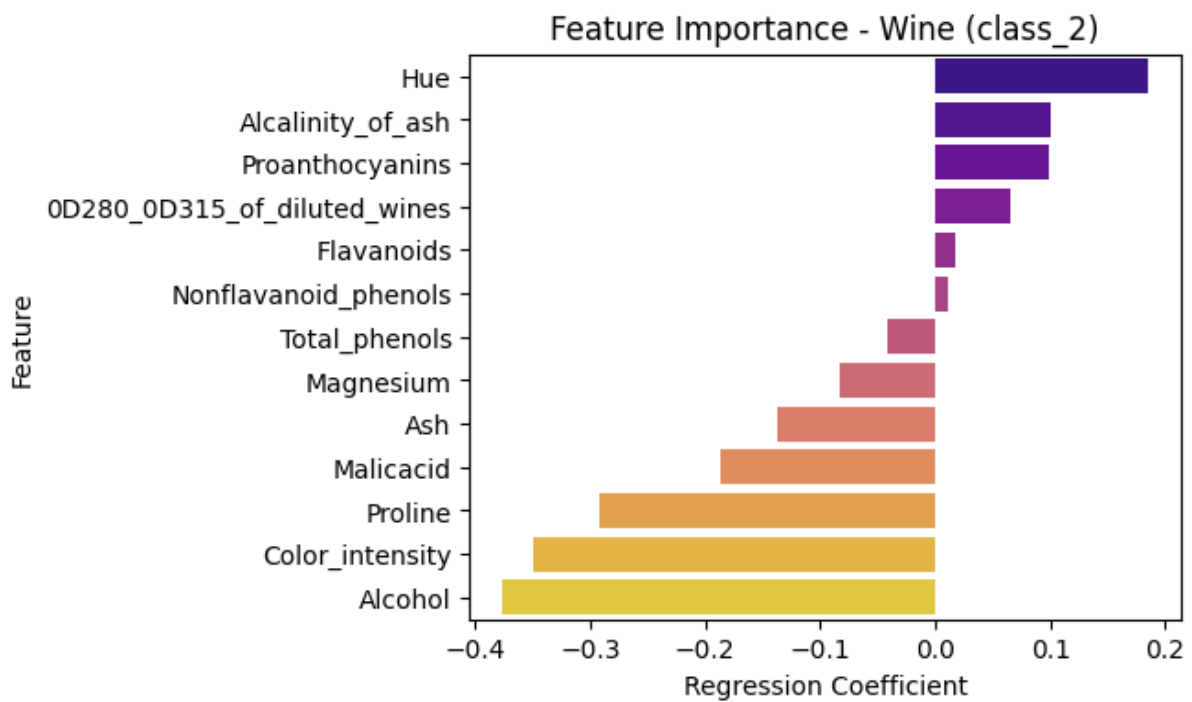


Figure 7: Bar Plot for feature importance for class 2 of wine dataset

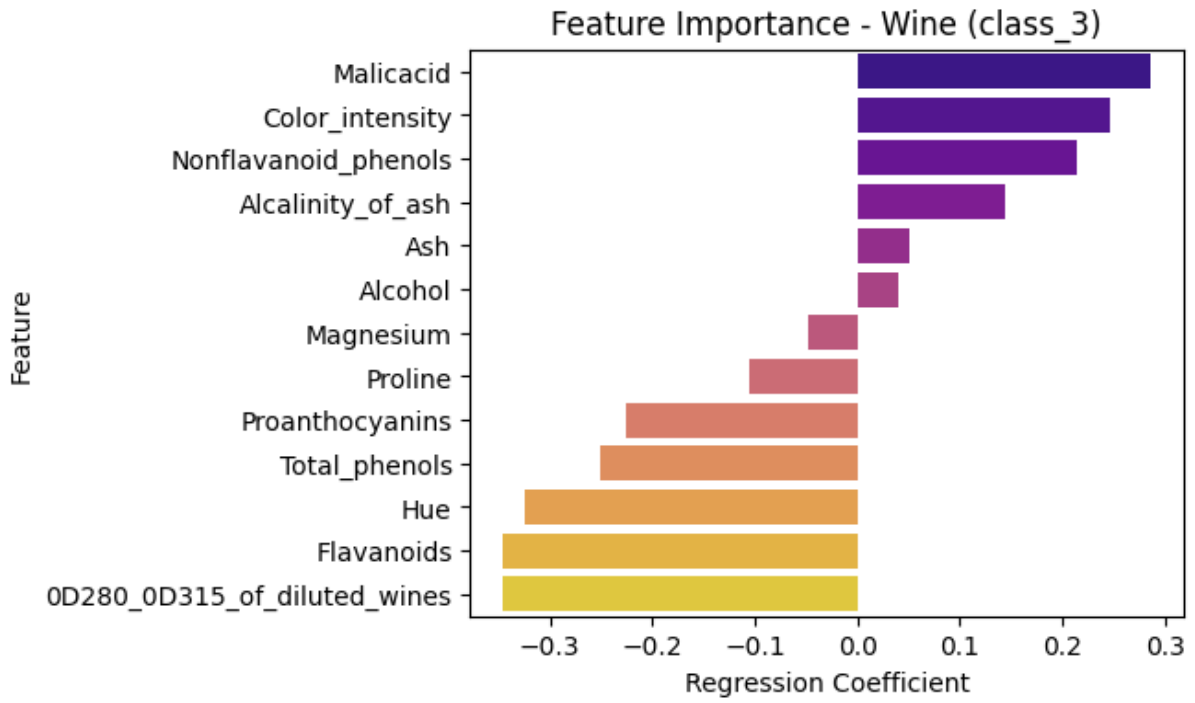


Figure 8: Bar Plot for feature importance for class 3 of wine dataset

- The difference between the analytical and approximate gradients was overall quite small, consistently around 2×10^{-16} .
- Convergence plot on how the logistic and multiclass classification converge given learning rate 0.005:

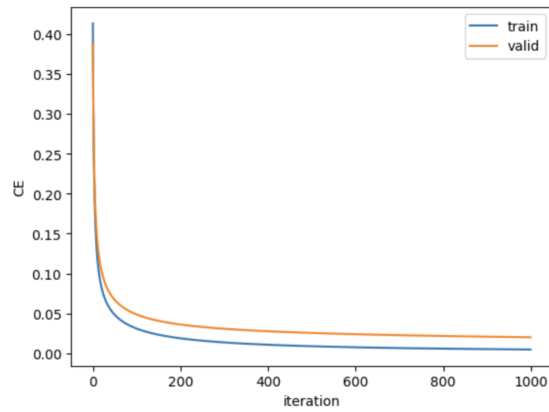


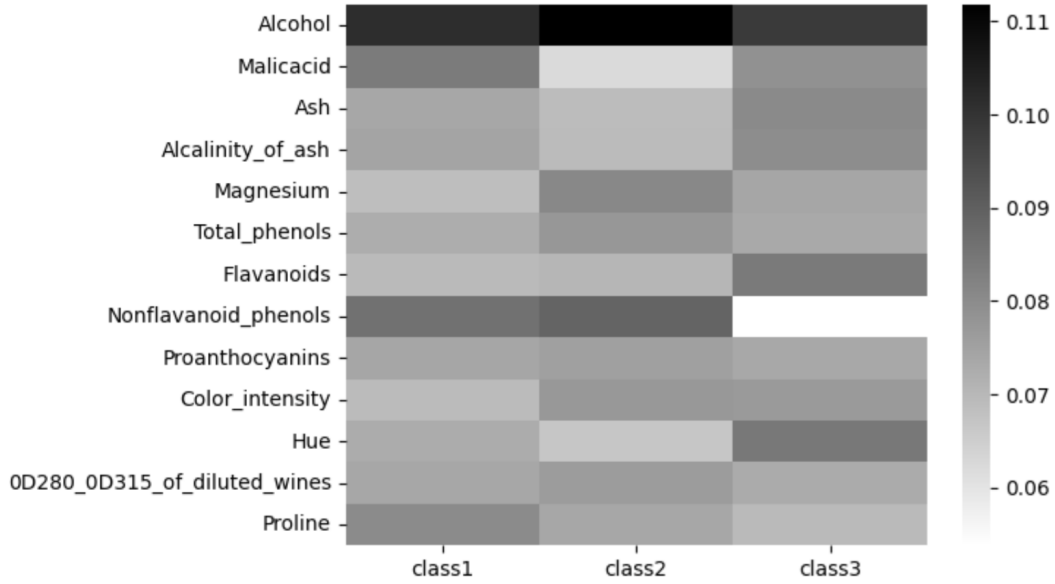
Figure 9: CE loss as a function of the number of iterations

- The classification accuracies for multiclass classification and for multivariate multiple regression are as follows:

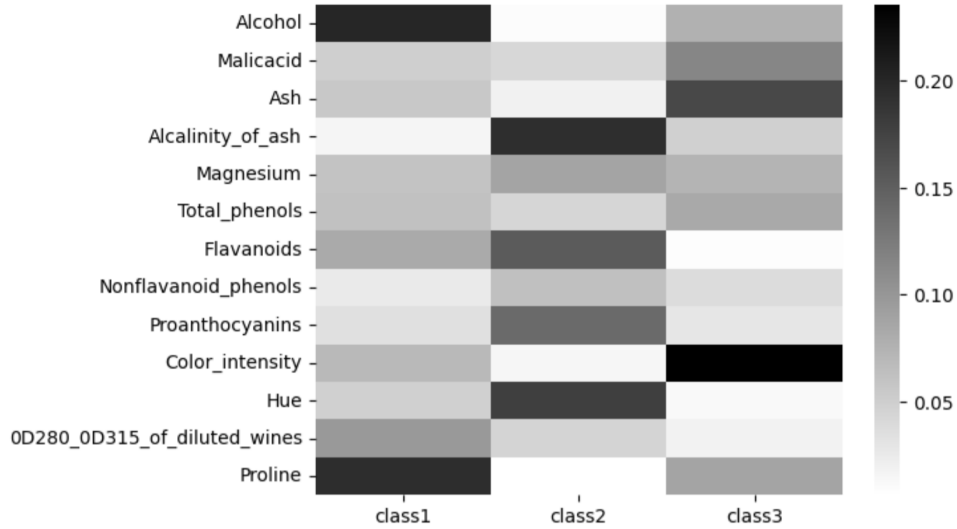
Multivariate Multiple Regression	Multi-Class classification
1.0 (train)	1.0 (train)
0.956 (validation)	1.0 (validation)
0.977 (test)	0.977 (test)

Tabel 1: Average correlation of different dataset types with the target variable for heart dataset.

- Heatmaps for the relationship between features (rows) and classes (columns):



Figuur 10: Heatmap for multivariate multiple regression



Figuur 11: Heatmap for multi-class classification

5 Discussion and Conclusion

For Binary Classification (Breast Cancer Dataset): For binary classification, we compared logistic regression and linear regression, evaluating their ROC curves, AUC scores, and feature importance rankings. The ROC curve shows that both models achieve high AUC values (0.99), indicating strong performance. However, logistic regression is more appropriate for classification since it directly models the probability of class membership, whereas linear regression treats the problem as a continuous prediction task, which may lead to poor probability estimates.

From the feature importance comparison, the ranking of features differs between the two models. In linear regression, the feature weights are derived from a least squares optimization, meaning they contribute linearly to the output. In contrast, logistic regression uses a logit transformation, which affects how features influence class probabilities. This difference explains why some features are ranked higher in linear regression but have less impact in logistic regression. For instance, features such as perimeter, concave points, and radius appear among the top-ranked predictors in both models, but their relative

impact varies between the two approaches.

The logistic regression coefficients are more concentrated around key discriminative features, suggesting that it focuses on variables that maximize class separation. In contrast, the linear regression coefficients (purple-yellow bar plot) are spread more evenly, meaning it considers all features to some extent, even if they have less predictive power. This difference highlights why cross-entropy loss (used in logistic regression) is more suitable for classification than MSE (used in linear regression), which assumes a continuous relationship that does not exist in classification tasks.

As an extra, we computed the ROC curve for KNN and DT as well. KNN achieved a near-perfect AUC (1.0), while the Decision Tree classifier lags behind (AUC = 0.89). This suggests that KNN and logistic regression generalize well, while decision trees may be prone to overfitting.

Finally, examining the top features, we see that attributes related to tumor shape and structure (e.g., perimeter, concavity, and smoothness) play an important role in distinguishing malignant from benign cases. These findings align with medical knowledge, where irregular shapes and more pronounced edges are indicators of malignancy.

For Multi-Classification (Wine Dataset): The results indicate that logistic regression achieves perfect accuracy (1.0) on both the training and validation sets, while multivariate regression achieves slightly lower validation accuracy (0.956). However, both models perform similarly on the test set with an accuracy of 0.977, indicating strong generalization.

The difference in performance can be attributed to the nature of the problem. Linear regression is designed for continuous outputs and minimizes Mean Squared Error (MSE), making it less suited for classification tasks where discrete class labels are needed. In contrast, logistic regression optimizes cross-entropy loss, which is specifically designed to distinguish between categorical classes, making it the more appropriate choice for classification. Additionally, the convergence plot demonstrates that logistic regression reaches lower cross-entropy values, suggesting a more effective optimization process.

To continue, by looking at our two heatmaps, we can find valuable findings about the importance of each feature for predicting different classes in multi-class logistic regression and multiple linear regression.

- **Feature Importance Differences:** The logistic regression heatmap shows higher contrast, indicating that some features contribute more significantly to specific classes. The linear regression heatmap has a more uniform distribution, meaning all features have more balanced weights. Logistic regression focuses on separating discrete classes using probabilistic decision boundaries. Some features become dominant in classifying a particular class. Linear regression tries to model a continuous relationship, assuming all features contribute linearly to the output. This leads to a more uniform feature importance distribution.
- **Top Features Per Class Change** In logistic regression, features like Alcohol, Color Intensity, proline and Flavanoids show higher importance for specific classes. In linear regression, Alcohol and flavanoids are also important factors. Features like Alcohol and Color Intensity in logistic regression likely have stronger discriminative power between classes. In contrast, linear regression does not differentiate classes well and treats the data as a regression problem rather than classification, leading to a different feature ranking.

6 Group Member Contributions

Negar was responsible for data acquisition, preprocessing, and feature analysis, as well as implementing the linear regression models. Deniz worked on the binary logistic regression, and Victoria handled the multiclass logistic regression. Deniz and Victoria also created the visualization tools (graphs and plots). All team members collaborated on reviewing and writing the report.

7 References

1. W. H. Wolberg, W. N. Street, and O. L. Mangasarian. *Machine learning techniques to diagnose breast cancer from fine-needle aspirates*. Cancer Letters, 77(2–3):163–171, 1994.
2. S. Aeberhard, D. Coomans, and O. de Vel. *Comparative analysis of statistical pattern recognition methods in high dimensional settings*. Pattern Recognition, 27(12):1415–1426, 1994.